

Management Practices and Mergers and Acquisitions

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Abstract. Using a novel data set of establishment-level management practices from the U.S. Census Bureau, we show that firms with more specific, formal, frequent, or explicit (i.e., “structured”) management practices tend to acquire establishments with less structured management practices and, following the acquisition, adopt more structured practices at the target establishments. These changes are larger when acquirers have a greater incentive and ability to make changes and are also associated with improvements in establishment performance. Overall, our findings suggest that the adoption of more structured management practices constitutes an important source of value creation in mergers and acquisitions.

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1. Introduction

How do mergers and acquisitions (M&A) create value? Synergies are the leading motive behind mergers. However, direct empirical evidence on the sources of these gains is limited. An often-overlooked source of synergies arises from improvements in management practices. Whereas business schools, the popular press, and research have long emphasized the importance of management, there is little empirical evidence on whether changes in management practices are a source of gains from mergers and acquisitions. A major challenge in addressing this question has been the absence of high-quality data on management practices.

In this study, we utilize a novel survey data set of establishment-level management practices from the U.S. Census Bureau to peek inside the “black box” of mergers and examine whether mergers create value through implementing more structured management practices.¹ Structured management practices are those that are more specific, formal, frequent, or explicit. Examples of more structured management practices include having more key performance indicators, clearly displayed production targets, and compensation more explicitly tied to a worker’s performance. Importantly, firms and establishments with more structured management practices perform better on metrics such as productivity, profitability, growth, and longevity (e.g., Bloom and Van Reenen 2007, Bloom et al. 2013a, Sadun et al. 2017). This performance persists over time, suggesting that more structured management practices can be a source of competitive advantages.

In the context of mergers, implementing more structured management practices at acquired establishments can be especially valuable, because it allows acquirers to better track performance and incentivize workers during the integration phase.²

Our data come from the 2010 Management and Organizational Practices Survey (MOPS), which is the first large-scale management practices survey created and introduced by Bloom et al. (2013b) and covers about 37,000 U.S. manufacturing establishments. The survey groups management practices into three categories: monitoring (six practices), production targets (two practices), and incentives (eight practices). Questions on monitoring practices focus on the extent to which performance is tracked and reviewed and whether these data are used to improve performance. More structured monitoring practices can improve the production process through enhanced collection and use of data. The questions on production target practices focus on the horizon and realism of the targets. Questions on incentive practices include promotion criteria, how pay and bonuses are distributed, and when bad performers are fired, where more structured practices are those that give strong rewards to employees based on their ability and effort. Whereas more structured production target and incentive practices do not necessarily impact productivity directly, they can improve performance by impacting employee decision making and effort. In addition, structured incentive practices can affect productivity through employee selection, whereby high-powered

incentives attract workers with high ability (e.g., Lazear 2000, Oyer and Schaefer 2005).

To utilize these 16 survey questions in our empirical tests, we follow prior work and create a single aggregate, normalized establishment-level management practices score that reflects all three of these categories (e.g., Bloom et al. 2013b, Bloom et al. 2019). Whereas the management practices score gives each survey question an equal weight, this score captures meaningful and value-relevant content and not just statistical noise. Firms and establishments with higher management scores have higher market valuations, profitability, productivity, value added, growth, survival rates, and innovation (Bloom et al. 2013b, Bloom et al. 2019).

To examine the effect of mergers on acquired establishments' management practices, we match the MOPS data to the Census Bureau's Longitudinal Business Database (LBD). The LBD provides annual coverage on employment and payroll information for the universe of all business establishments with at least one paid employee in the United States. The LBD includes unique firm- and establishment-level identifiers, making it possible to follow an establishment as it changes owners. We are therefore able to track changes in the management practices of targets and acquirers separately before and after each merger event. Moreover, our data cover both public and private firms, which allows us to observe both partial- and whole-firm acquisitions. In our sample, about 5% of acquisitions involve acquiring more than 80% of a firm's establishments, and about 50% of acquisitions are of one or two of a firm's establishments. Thus, nearly all of the acquisitions in our sample are partial acquisitions. We begin our analysis by examining whether there is a relation between management practices and the likelihood that a firm becomes an acquirer and an establishment becomes a target.³ Overall, we find strong evidence that firms with more structured management practices are more likely to become acquirers and some evidence that establishments with less structured management practices are more likely to be acquired. In addition, using a regression specification that resembles gravity models used to study trade flows between countries, we show that firms with more structured management practices tend to acquire establishments with relatively less structured management practices.

We then turn to our main analyses. The 2010 MOPS asks respondents questions about their establishment's management practices in 2010 and 2005. This longitudinal component of the survey allows us to implement a difference-in-differences empirical specification to examine the effect of acquisitions on management practices. Using this research design, we document that, relative to a control group of establishments that did not get acquired, acquired establishments adopt more

structured management practices. Further, this effect is not concentrated in one particular category of management practices, as we find that monitoring, production target, and incentive practices separately become more structured. We also show that, in addition to the management practices at acquired establishments becoming more structured on average, the target's management practices converge to those of the acquirer's incumbent establishments. Using a measure of cosine similarity to capture the distance between a target's and an acquirer's management practices, we also find that the management practices of targets look more similar to those of the acquirer after the acquisition, suggesting that the acquirer applies its own management style to the newly acquired establishment.

We implement our difference-in-differences research design in a change analysis and therefore control for any time-invariant firm- and establishment-level characteristics. In these change regressions, we also include industry and state fixed effects. Because the data set has only one cross section in changes, these fixed effects control for any industry and state characteristics that changed between 2005 and 2010 that could be correlated with changes in an establishment's management practices and the likelihood that it is acquired. As a result, the primary identification threat that would not allow us to attribute the adoption of more structured management practices to the acquisition event is if these changes are a continuation of a pre-existing trend. For example, if acquirers tend to purchase establishments that are already adopting more structured management practices, then the change we observe could simply be a continuation of this trend and not a result of the acquisition. However, inconsistent with this alternative explanation, we do not find a relation between changes in management practices and the likelihood that an establishment is acquired in an out-of-sample test. We also conduct a test in which we limit the sample to only targeted establishments, creating a treatment group consisting of completed announced acquisitions and a control group consisting of withdrawn announced acquisitions (e.g., Seru 2014). Consistent with the acquisition event driving our findings, we find that, compared with the change in the management practices of targets in withdrawn bids, target establishments in completed deals adopt more structured management practices.

In our next set of tests, we explore cross-sectional variation in the effect of acquisitions on an establishment's management practices. If our results are driven by acquirers adopting more structured management practices at target establishments to realize synergies, then we should find that these changes are greater when acquirers have a greater incentive and ability to do so. We test this prediction in four ways. First, we capture an acquirer's comparative advantage in management practices using its management

score before the acquisition. Acquirers that already have more structured management practices have the skills, knowledge, and resources needed to realize greater synergies by applying their management styles to targets. Second, we measure an acquirer's ability to influence organizational change with the extent to which managers have decision-making authority. When decision control is concentrated with managers, they are able to make decisions and implement changes faster (e.g., Wally and Baum 1994, Baum and Wally 2003). Third, we use the extent of product market competition in the establishment's industry to capture instances when acquirers have a greater incentive to adopt more structured management practices. More intense product market competition serves as a disciplinary device that mitigates managerial slack (e.g., Giroud and Mueller 2010), giving acquirers greater incentive to improve economic efficiency. Fourth, acquired establishments are more likely to adopt more structured management practices when targets and acquirers have complementary operations, because the acquirer already possesses a comparative advantage and expertise that it can use to manage the target's operations (e.g., Maksimovic and Phillips 2001, Frésard et al. 2017). Consistent with these predictions, we find that acquired establishments adopt even more structured management practices when (i) acquirers start with more structured practices, (ii) there are fewer layers of direct reports from the factory floor to the establishment manager, (iii) managers are primarily the ones prioritizing and allocating tasks to production workers, (iv) establishments are in more competitive industries, and (v) the acquirer and target operate in the same industry.

Next, we examine whether the adoption of more structured management practices at acquired establishments translates into improved performance, to test whether these changes are a source of value creation in mergers. We capture establishment performance using several measures, including total factor productivity, value added per employee, value added per production hour, profit margins, and return on capital. Consistent with the notion that adopting more structured management practices is a source of gains from M&A, we find that postacquisition improvements in performance are larger when acquired establishments adopt more structured management practices.⁴

We also conduct a number of robustness tests. Our finding that establishments adopt more structured management practices following an acquisition is robust to controlling for possible recall bias introduced by the nature of the longitudinal component of the MOPS. The results also hold after controlling for survey respondent position fixed effects. Potential biases due to survey respondents manipulating their responses also do not appear to drive our findings.

Our study complements the existing literature in at least three dimensions. First, we contribute to work on the sources of gains from M&A. In theory, such value creation could arise from many sources, including economies of scale or scope, increases in managerial efficiency, improvements in production techniques, or increases in market power (e.g., Trautwein 1990, Andrade et al. 2001). Although it is well documented that successful mergers often increase the value of the combined company (e.g., Healy et al. 1992, Betton et al. 2008), direct empirical evidence showing how these gains occur has historically been limited.

Recently, hand collection and better access to confidential and proprietary data have led to work that investigates a number of other specific channels of the sources of synergies from M&A. Gains can come from improved cash flows and productivity resulting from better resource allocation and product differentiation (e.g., Lichtenberg and Siegel 1987, Healy et al. 1992, McGuckin and Nguyen 1995, Maksimovic and Phillips 2001, Schoar 2002, Hoberg and Phillips 2010, Maksimovic et al. 2011, Li 2013). Gains can also arise from interest tax shields, improvements in product quality, and efficiency-enhancing cost savings resulting from economies in capital expenditures, investments in working capital, and advertising (e.g., Devos et al. 2009, Fee et al. 2012, Sheen 2014). In the case of horizontal mergers, gains could come from enhanced buying and pricing power (e.g., Kim and Singal 1993, Focarelli and Panetta 2003, Fee and Thomas 2004, Ashenfelter and Hosken 2010, Bhattacharyya and Nain 2011).⁵

Our paper is most closely related to work on the role of managers and organizational capital in M&A deal success. Early work proxies for managerial performance using Tobin's Q and finds that gains from M&A are largest when high-Q firms acquire low-Q firms (Lang et al. 1989, Servaes 1991). Consistent with these findings, Wang and Xie (2009) and Masulis et al. (2007) use corporate governance metrics related to antitakeover provisions to capture how well a firm is managed and find larger M&A gains when firms with stronger shareholder rights acquire firms with weaker shareholder rights. To the extent that having more structured management practices can be a source of organizational capital, our study is also related to that of Li et al. (2018, pp. 1871–1872), who define organizational capital as “the body of knowledge and business processes and systems that facilitates the match between labor and physical production facilities and allows firms to utilize resources more efficiently.” They find that higher acquirer organizational capital, proxied with capitalized selling, general, and administrative expenses, is associated with better M&A performance. Whereas all of these studies suggest that managers and oversight matter for M&A success, inferences are limited due to the use

of indirect proxies. For instance, Tobin's Q could reflect factors other than managerial performance, such as agency problems, investment opportunities, and market misvaluation. By using detailed data on establishment-level management practices, we overcome some of these limitations and show how specific practices change after an acquisition and that these changes are associated with improvements in performance.

Second, our study extends work that links management practices to productivity (e.g., Ichniowski et al. 1997, Lazear 2000, Black and Lynch 2001). In particular, more structured and better management practices lead to improvements in quality, inventory, and output, which translate into higher productivity (Bloom et al. 2013a, Bender et al. 2018). Differences in management practices across countries can also in part explain cross-country differences in productivity (Bloom and Van Reenen 2007, Bloom et al. 2010).

Last, our results relate to recent work on how culture and other characteristics of merged companies converge, and to work on how cultural fit, human capital relatedness, and technological overlap affect merger outcomes (e.g., Atalay et al. 2014, Bena and Li 2014, Bereskin et al. 2018, Lee et al. 2018, Li et al. 2021).

2. Databases and Sample Selection

2.1. The Management and Organizational Practices Survey

Our source of establishment-level data on management practices comes from the U.S. Census Bureau's 2010 Management and Organizational Practices Survey (MOPS). The 2010 MOPS is a supplement to the 2010 Annual Survey of Manufacturers (ASM), which covers approximately 70% of U.S. manufacturing activities in terms of total value of shipments. We describe the construction of the MOPS and how it was distributed in detail in the online appendix. For the ASM, reporting is mandatory for establishments selected to participate in the survey. Establishments must report all of the data to the government by law, and fines are levied for misreporting.

The purpose of the MOPS is to better understand current and evolving management and organizational practices and to assist in identifying determinants of establishment productivity and growth. The 2010 survey consists of 37 questions in three sections: management practices, organizational structure, and background characteristics. Following prior work, we focus on the first 16 questions on management practices (Appendix A lists these questions). These 16 questions fall into three categories: monitoring (six practices), production targets (two practices), and incentives (eight practices). These questions are based on the survey questions of management practices used in Bloom and Van Reenen (2007) and the World Management Survey, which are based in part on the

principles of continuous monitoring, evaluation, and improvement from lean manufacturing (e.g., Womack et al. 1990).

The six questions on monitoring ask the extent to which establishments track and review performance and whether these data are used to improve performance. For example, one question is "How many key performance indicators were monitored at this establishment?" The two questions on production targets attempt to assess how well establishments set production goals. For instance, one question is, "What best describes the time frame of production targets at this establishment?" The eight questions on incentives examine how employees are promoted, rewarded, and retained, or alternately reprimanded and dismissed. For example, one question is, "What were nonmanagers' performance bonuses usually based on?"

The 2010 MOPS was distributed as a supplement of the ASM mailout sample. This portion of the survey comprises a probability sample of approximately 50,000 manufacturing establishments from a universe of approximately 101,250 establishments. The survey was sent electronically and by mail to the ASM respondent for each establishment, and the majority of these respondents were establishment managers (63.9%), divisional managers (14.9%), or chief executive officers (CEOs) (8.8%). On average, respondents have worked at their establishment for 11.7 years at the time of the survey. The 2010 MOPS received responses from approximately 37,000 establishments (about 78% of the establishments to whom the survey was successfully delivered). To create a longitudinal data set, each respondent was asked about her or his establishment's management practices in 2005.⁶ This longitudinal component of the survey allows us to observe an establishment's management practices in 2005 and 2010, allowing us to calculate changes in its management practices over this five-year period.

Following prior work (Bloom et al. 2013b, Bloom et al. 2019), our primary measure of establishment-level management practices is an aggregate, normalized management score that captures all three categories of management practices. The creation of this measure involves two steps. First, we normalize each of the responses to the 16 management practices questions to a value between zero and one, with the responses associated with the most structured management practices receiving a value of one (see Appendix A for exact scoring of each question). In general, more structured management practices are those that are more specific, formal, frequent, or explicit. For example, responses to the question "What best describes what happened at this establishment when a problem in the production process arose?" are (i) no action was taken, (ii) we fixed it but did not take further action, (iii) we fixed it and took action to

make sure that it did not happen again, and (iv) we fixed it and took action to make sure that it did not happen again, and had a continuous improvement process to anticipate problems like these in advance. Converting these responses to our 0-1 scale, responses (i) to (iv) receive values of 0.0, 0.333, 0.667, and 1.0, respectively. Similarly, for the question “When was an under-performing non-manager reassigned or dismissed?” possible responses include (i) rarely or never, (ii) after six months of identifying nonmanager under-performance, and (iii) within six months of identifying nonmanager under-performance. Converting these responses to our 0-1 scale, responses (i) to (iii) receive values of 0.0, 0.5, and 1.0, respectively. Second, we take the simple average of the 16 normalized responses to obtain each establishment’s management score.⁷

Consistent with this management score derived from the MOPS capturing meaningful and value-relevant management practices, higher values are associated with numerous metrics of better establishment performance (e.g., Bloom et al. 2013b, Bloom et al. 2019). In addition, higher values of similar management scores derived from the World Management Survey, which the MOPS is based on, are also correlated with better firm performance across a wide range of industries and countries (e.g., Bloom and Van Reenen 2007, Bloom et al. 2013a, Bloom et al. 2015).

2.2. Changes in Ownership

Central to our analyses is the ability to accurately identify all forms of partial- and whole-firm acquisitions. We identify ownership changes of establishments by tracking the same establishment over time using the Census Bureau’s Longitudinal Business Database (LBD). The LBD provides annual coverage for all business establishments in the United States with at least one paid employee and is available from 1976 to 2014.

The LBD contains unique establishment- and firm-level identifiers, as well as employment, payroll, industry, and detailed geographical location information for each establishment. These unique establishment- and firm-level identifiers allow us to follow an establishment as its owners change over time. Thus, we are able to identify years when the ownership of each establishment changes from one firm to another. In addition, these establishment-level data allow us to track changes in the management practices and performance of targets and acquirers separately before and after each merger event, even after a public establishment is acquired by a private firm. This feature is very attractive, as it allows us to observe changes in the management practices of each establishment, which is an improvement over studies that rely on aggregate firm-level accounting or stock market data to gauge the impact of M&A.

2.3. Sample Selection

Our final sample consists of approximately 14,000 establishment-level observations in 2010.⁸ Our sample is smaller than the approximately 37,000 establishments that responded to the MOPS for the following reasons: (i) we require establishments to have data in the ASM and LBD in both 2005 and 2010,⁹ (ii) we exclude administrative records to minimize measurement error (e.g., Maksimovic et al. 2011, Giroud and Mueller 2015), (iii) we require each establishment to have strictly positive values of sales and employment, and (iv) we follow Bloom et al. (2013b) and require that respondents answer at least 11 of the 16 questions on management practices.

Panels A and B of Table 1 tabulate summary statistics for the main establishment- and firm-level variables used in our regressions, respectively. The overall management practices score in 2005 was 0.59 on average. Establishments had the most structured management practices related to production targets with a value of 0.67 and the least structured management practices related to monitoring with a value of 0.56. On average, management practices became more structured between 2005 and 2010, with an increase of 16.7% ($= 0.0986/0.591$), 29.7%, 14.7%, and 7.5% in the management practices scores of the overall, monitoring, production target, and incentive scores, respectively. Table A1 in the online appendix tabulates the mean and standard deviation for each of the 16 question’s scores for 2005 and 2010. The score decreased for only one question, decreasing from 0.38 to 0.36 for question 11, “What were managers’ performance bonuses usually based on?” Table OA.2 depicts a transition matrix showing what fraction of establishments changed their responses to the management practices questions from 2005 to 2010 and by how much. For about three-fourths of the questions, firms kept the same management practices or moved their response up one level.

Of the 14,000 establishments, about 14.0% experience an ownership change between 2006 and 2010. On average, establishments downsized over this period and mostly in terms of employment, with the number of employees shrinking by 16.4% and capital stock shrinking by less than 1%. This latter result is not surprising and is consistent with Giroud and Mueller (2017), as our sample starts during an economic peak two years before the Great Recession in 2007–2008 and ends a year after the recession’s trough in 2009.

Table 2 compares the 2005 values of establishment characteristics across four separate groups: (i) establishments that are eventually acquired, (ii) establishments that are not acquired but belong to a firm that had at least one establishment acquired, (iii) establishments that belong to a firm that made an acquisition, and (iv) establishments that are not acquired and belong to a firm

Table 1. Summary Statistics

Panel A: Establishment-level summary statistics (14,000 observations)

	Mean	Standard deviation
Δ MP Score	0.0986	0.1305
Pos. Δ MP Score	0.7321	0.4429
Neg. Δ MP Score	0.1161	0.3203
MP Score ₀₅	0.5914	0.1480
Δ MP Monitoring Score	0.1669	0.1831
MP Monitoring Score ₀₅	0.5628	0.1933
Δ MP Target Score	0.0979	0.2346
MP Target Score ₀₅	0.6682	0.2494
Δ MP Incentive Score	0.0441	0.0441
MP Incentive Score ₀₅	0.5902	0.1851
Ownership Change	0.1395	0.3464
Δ Ln(Employment)	−0.1640	0.4950
Δ Ln(Capital Stock)	−0.0011	0.3370
Δ Managerial Education	0.2883	0.6870
Δ Manager-Employee Ratio	0.0032	0.4333
Δ Unionization Rate	−0.0179	0.4280
Age ₀₅	26.410	9.8790
Ln(Employment) ₀₅	5.0600	1.1350
Ln(# Parent Estabs.) ₀₅	3.4750	1.7590

Panel B: Firm-level summary statistics (5,000 observations)

	Mean	Standard deviation
MP Score ₀₅	0.6550	0.1340
Employment ₀₅	718.30	2291.0
Age ₀₅	28.040	9.2400
TFP ₀₅	1.7780	0.6225

Notes. This table reports summary statistics for the main variables in the regression models. Panel A presents summary statistics for key establishment-level variables (approximately 14,000 observations). Panel B presents summary statistics for key firm-level variables (approximately 5,000 observations). Δ indicates that variables are measured in changes from 2005 to 2010. *MP Score* is the establishment's normalized management score. *Pos. Δ MP Score* (*Neg. Δ MP Score*) is an indicator variable that is set to one if the change in an establishment's normalized management score between 2005 and 2010 is strictly positive (negative) and zero otherwise. *MP Monitoring Score*, *MP Target Score*, and *MP Incentive Score* are the normalized management scores based on the questions on monitoring, production target, and incentive practices, respectively. *Ownership Change* is an indicator variable that is set to one if an establishment is acquired between 2006 and 2010 and zero otherwise. In panel B, establishment-level variables are aggregated up to the firm level. Appendix B provides further definitions of variables.

that did not make an acquisition. Focusing on the acquired and nonacquired establishments of a target firm (columns 1 and 2), the results show that acquired establishments tend to have lower management scores, fewer employees and physical assets, less-educated managers, and lower union membership rates. Acquired establishments are also younger and less productive. Ideally, establishments that are acquired and those that are not acquired would be similar along each dimension. However, because they are not, we control for each variable in our main regressions to account for these differences. Comparing acquired establishments to those of acquiring firms (columns 1 and 3), the results show that acquiring-firm establishments have

higher management scores, more physical assets, and more educated managers. Acquiring-firm establishments are also older and more productive.

Table 3 shows the average management score in 2005 for the 21 three-digit North American Industry Classification System (NAICS) manufacturing industry codes in our sample. Overall, it does not appear that there are substantial differences in the management practices of establishments in different industries. The industry with the most structured management practices is "Transportation Equipment Manufacturing," with an average management score of 0.624. The industry with the least structured management practices is "Leather and Allied Product Manufacturing," with an average management score of 0.541. To put this difference into perspective, the standard deviation of management scores across all establishments is 0.148.

3. Empirical Results

3.1. Management Practices and the Likelihood of Becoming an Acquirer and a Target

Do management practices predict whether a firm becomes an acquirer and an establishment becomes a target? We first use firm-level data to estimate the likelihood that a firm becomes an acquirer in any of the years between 2006 and 2010 as a function of its level of management practices in 2005. In this firm-level analysis, we calculate one firm-level variable for each characteristic by taking the employment-based weighted average of each characteristic across all of the firm's establishments.

The dependent variable is either an indicator variable that is set to one if a firm acquires an establishment between 2006 and 2010 and zero otherwise, or the natural logarithm of one plus the number of acquisitions the firm makes between 2006 and 2010. We estimate ordinary least-squares (OLS) regressions and control for the 2005 values of the following firm-level variables: total number of employees, total factor productivity, number of establishments owned by the firm, capital stock, the average education level of managers, the ratio of the number of managers to non-managerial employees, and the union membership rate. In this test, we also include industry fixed effects based on six-digit NAICS industry codes and state fixed effects to control for different industry and state characteristics between firms that could be correlated with the level of a firm's management score and the likelihood that it becomes an acquirer. We cluster standard errors by firm.

The results in columns 1 and 2 of Table 4 show a positive and statistically significant relation between a firm's management score and the likelihood that it becomes an acquirer, as well as the number of acquisitions it makes. Thus, firms with more structured management practices are more likely to participate

Table 2. Comparison of Means Across Groups of Establishments

	Acquired establishments	Establishments not acquired but belong to a firm that had an acquired establishment	Establishments of acquiring firms	All other establishments	Difference	Difference
	(1)	(2)	(3)	(4)	(1) vs. (2)	(1) vs. (3)
<i>MP Score</i> ₀₅	0.5813	0.5953	0.6224	0.5755	−0.0140**	−0.0411***
<i>Monitoring Score</i> ₀₅	0.5571	0.5745	0.5978	0.5426	−0.0174**	−0.0407***
<i>Target Score</i> ₀₅	0.6594	0.6856	0.6976	0.6532	−0.0262**	−0.0382***
<i>Incentive Score</i> ₀₅	0.5769	0.5852	0.6189	0.5776	−0.0083	−0.042***
<i>Ln(Employment)</i> ₀₅	5.187	5.299	5.214	4.908	−0.112**	−0.027
<i>Ln(Capital Stock)</i> ₀₅	9.876	10.24	10.01	9.407	−0.364***	−0.134***
<i>Managerial Education</i> ₀₅	2.812	3.157	2.953	2.474	−0.345***	−0.141***
<i>Manager-Employee Ratio</i> ₀₅	0.0893	0.1003	0.0912	0.0987	−0.011	−0.0019
<i>Unionization Rate</i> ₀₅	2.363	2.568	2.326	1.911	−0.205***	0.037
<i>Age</i> ₀₅	21.64	23.23	22.69	20.38	−1.59***	−1.05***
<i>TFP</i> ₀₅	−0.0158	0.0337	0.0152	−0.0085	−0.0495***	−0.0310***

Notes. This table compares mean values for the main variables in the regression models across different groups of establishments. Column 1 reports means for establishments that are acquired. Column 2 reports means for establishments that are not acquired but that belong to a firm that had at least one of its establishments acquired. Column 3 reports means for establishments that belong to a firm that made an acquisition. Column 4 reports means for all other establishments, which includes establishments of firms that did not make an acquisition nor had an establishment acquired. *, **, and *** in the columns labeled “Difference” indicate significance at the 10%, 5%, and 1% levels, respectively, for a *t*-test of whether the two samples have equal means. *MP Score*, *MP Monitoring Score*, *MP Target Score*, and *MP Incentive Score* are the normalized management scores based on all of the management practices questions and the questions on monitoring, production target, and incentive practices, respectively. Appendix B provides further definitions of variables.

in the acquisition market. For example, the coefficient estimates in column 1 imply that a one-standard-deviation-higher management score is associated with a 0.77 ($= 0.0573 \times 0.1340$) percentage-point-higher likelihood that a firm becomes an acquirer. Given that 10.3% of firms eventually become an acquirer in

our sample, a firm with a one-standard-deviation-higher management score is 7.5% more likely to become an acquirer. Because we only observe the management practices of manufacturing establishments, the average management score across a firm’s establishment might not be representative for conglomerates.

Table 3. Average Management Score by NAICS Industry

Three-digit NAICS industry	<i>Avg. MP Score</i> ₀₅
336 - Transportation Equipment Manufacturing	0.6235
331 - Primary Metal Manufacturing	0.6062
334 - Computer and Electrical Manufacturing	0.6053
322 - Paper Manufacturing	0.6049
325 - Chemical Manufacturing	0.6047
312 - Beverage and Tobacco Product Manufacturing	0.6001
339 - Miscellaneous Manufacturing	0.5973
326 - Plastics and Rubber Products Manufacturing	0.5951
313 - Textile Mills	0.5936
321 - Wood Product Description	0.5923
335 - Electric Equipment, Appliance, and Component Manufacturing	0.5923
311 - Food Manufacturing	0.5918
324 - Petroleum and Cool Products Manufacturing	0.5844
315 - Apparel Manufacturing	0.5808
337 - Furniture and Related Product Manufacturing	0.5797
327 - Nonmetallic Mineral Product Manufacturing	0.5774
332 - Fabricated Metal Product Manufacturing	0.5746
333 - Machinery Manufacturing	0.5733
314 - Textile Product Mills	0.5713
323 - Printing and Related Support Activities	0.5639
316 - Leather and Allied Product Manufacturing	0.5409

Notes. This table reports the average management score in 2005 for the 21 three-digit NAICS manufacturing industries in our sample. *Avg. MP Score*₀₅ is the mean normalized 2005 management score across all firms in a three-digit NAICS industry.

Table 4. Management Practices and the Likelihood of Becoming an Acquirer or a Target

	<i>Became Acquirer</i>	<i>Ln(# Acqs)</i>	<i>Became Acquirer</i>	<i>Became Acquirer</i>	<i>Became Target</i>	<i>Became Target</i>
	(1)	(2)	(3)	(4)	(5)	(6)
<i>MP Score</i> ₀₅	0.0573*** (0.0165)	0.0528*** (0.0182)	0.0695*** (0.0154)	0.0549** (0.0269)	−0.0261* (0.0157)	−0.0284* (0.0163)
<i>Ln(Employment)</i> ₀₅	0.0646*** (0.0046)	0.0795*** (0.0051)	0.0313*** (0.0076)	0.0667*** (0.0046)	0.0039 (0.0037)	0.0057 (0.0037)
<i>TFP</i> ₀₅	0.0116* (0.0063)	0.0137** (0.0070)	0.001 (0.0086)	0.0122* (0.0063)	−0.0200* (0.0108)	−0.0195* (0.0108)
<i>Ln(# Parent Estabs)</i> ₀₅	0.0522*** (0.0043)	0.0535*** (0.0048)	0.0270*** (0.0045)	0.0458*** (0.0040)		
<i>Ln(Capital Stock)</i> ₀₅	0.0021 (0.0044)	−0.004 (0.0048)	0.0124** (0.0057)	0.0013 (0.0044)	−0.0106*** (0.0030)	−0.0107*** (0.0030)
<i>Managerial Education</i> ₀₅	0.0023 (0.0029)	0.0004 (0.0032)	0.0019 (0.0038)	0.0023 (0.0029)	−0.0039** (0.0016)	−0.0039** (0.0016)
<i>Manager-Employee Ratio</i> ₀₅	−0.0025 (0.0101)	−0.0047 (0.0111)	0.0021 (0.0126)	−0.0021 (0.0101)	0.0051** (0.0023)	0.0052** (0.0023)
<i>Unionization Rate</i> ₀₅	0.0019 (0.0026)	0.0035 (0.0028)	0.0036 (0.0037)	0.0018 (0.0026)	−0.0018 (0.0012)	−0.0018 (0.0012)
<i>Age</i> ₀₅					−0.0003 (0.0002)	−0.0003 (0.0002)
<i>Distance from HQ</i> ₀₅						0.0008 (0.0049)
<i>Core Industry</i> ₀₅						−0.0352*** (0.0085)
Industry FEs	Yes	Yes	Yes	Yes	No	No
State FEs	Yes	Yes	Yes	Yes	Yes	Yes
Firm FEs	No	No	No	No	Yes	Yes
Observations	5,000	5,000	2,000	5,000	14,000	14,000
Adjusted R ²	0.1502	0.1583	0.0431	0.1485	0.8218	0.8223

Notes. This table reports the results from OLS regressions examining the relation between management scores and the likelihood that a firm becomes an acquirer or that an establishment is acquired. The unit of observation in columns 1–4 is at the firm level. Column 3 restricts the sample to firms with at least 75% of its employees in the manufacturing sector. The dependent variable *Became Acquirer* in columns 1, 3, and 4 is an indicator variable that is set to one if a firm acquires an establishment between 2006 and 2010. The dependent variable *Ln(# Acqs)* in column 2 is one plus the natural logarithm of the number of establishments that a firm acquires between 2006 and 2010. All firm-level variables are calculated as the employment-based weighted average of the variable across all of a firm's establishments in 2005 (in column 4, we instead use the values of the firm's largest establishment). The unit of observation in columns 5 and 6 is at the establishment level, and the dependent variable *Became Target* is an indicator variable that is set to one if an establishment is acquired between 2006 and 2010. *Distance from HQ* is the natural logarithm of the number of miles between an establishment and the firm's headquarters. *Core Industry* is an indicator variable that equals one if the establishment is in the same six-digit NAICS industry as the firm's primary operating segment. Industry fixed effects (FEs) are based on six-digit NAICS industry codes. Appendix B provides further definitions of the variables. Standard errors in parentheses are heteroscedasticity-robust and clustered by firm. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

To address this concern, we repeat the analysis in column 1 but restrict the sample to firms that have at least 75% of their employees working for establishments in the manufacturing sector, and the results continue to hold (column 3). As a second robustness check, we repeat the analysis in column 1 but use the characteristics of a firm's largest establishment instead of an average across all of its establishments, and the results continue to hold (column 4).

Next, we use the establishment-level data to estimate the likelihood that an establishment is acquired in any of the years between 2006 and 2010 as a function of its level of management practices in 2005. We estimate OLS models and control for the same establishment-level controls as in columns 1–4 and an

establishment's age. In addition to including state fixed effects, we also include firm fixed effects to control for differences in characteristics between firms that could be correlated with the level of an establishment's management practices and the likelihood that it is acquired. Thus, the coefficient estimate on the management score represents the effect of an establishment's management practices relative to the average level of management practices of other establishments belonging to the same firm on the likelihood that the establishment is acquired. We continue to cluster standard errors by firm.

The results in column 5 show a negative and statistically significant relation between an establishment's management score and the likelihood that it is

acquired between 2006 and 2010. Hence, establishments with less structured management practices are more likely to be acquired. Given that 14.0% of establishments in our sample are eventually acquired, the coefficient estimates in column 5 imply that an establishment with a one-standard-deviation-lower management score is 2.8% more likely to become a target ($= 0.0261 \times 0.148/0.140$). Consistent with our univariate results in Table 2, the results also show that targets are more likely to sell less productive establishments, smaller establishments, and establishments with less-educated managers. In column 6, we repeat this analysis and include the distance between an establishment and the firm's headquarters and an indicator variable for whether the establishment is in the firm's core industry (six-digit NAICS industry code with the largest number of employees) as additional predictors. The results show that firms are also more likely to divest noncore establishments but distance from headquarters is not a factor.

Overall, the results in Table 4 are consistent with firms with more structured management practices acquiring establishments with less structured management practices. Another approach to test this conclusion is to estimate a specification that resembles gravity models used in economics to study trade flows between countries. For this test, we match each establishment that is eventually acquired to all potential acquiring firms that have management scores. Thus, each observation represents a target establishment-acquirer firm pair. We then estimate the following pooled linear probability model:

$$Target_{ij} = \beta_1 \Delta MPS_{ij} + \Gamma' \Delta X_{ij} + \eta_{ik,jk} + \nu_{is,js} + \varepsilon_{ij}, \quad (1)$$

where $Target_{ij}$ is an indicator variable that is set to one if establishment i is acquired by firm j . The variable ΔMPS_{ij} measures the difference between firm j 's management score in 2005 and establishment i 's management score in 2005. Similar to the prior analysis, firm j 's management score is the employee-based weighted-average management score across all of its establishments. A positive coefficient on β_1 would indicate that firms with more structured management practices tend to acquire establishments with relatively less structured management practices. The regression also includes the difference in control variables between firm j in 2005 and establishment i in 2005 (ΔX_{ij}). For this test, we control for differences in the following variables: number of employees, capital stock, average manager education, the ratio of managers to nonmanagerial employees, the union membership rate, and age. We also include acquirer firm-target establishment industry ($\eta_{ik,jk}$) and state ($\nu_{is,js}$) pair fixed effects. We cluster standard errors by target establishment.

Table 5 presents the results of this analysis. Consistent with the results in Table 4, we find that firms with higher management scores tend to acquire establishments with lower management scores. In terms of economic significance, the coefficient estimate on *Difference MP Score*₀₅ implies that establishment-firm pairs in which the potential acquiring firm has a one-standard-deviation-higher management score than the establishment increases the likelihood that the firm acquires the particular establishment by 14.0% relative to the unconditional mean.

3.2. Mergers and Changes in Management Practices

Do establishments adopt more structured management practices following a merger? To test this question,

Table 5. Gravity Model of the Likelihood of Becoming a Target

	Became Target (1)
<i>Difference MP Score</i> ₀₅	0.0753*** (0.0109)
<i>Difference Ln(Employment)</i> ₀₅	0.0008 (0.0022)
<i>Difference Ln(Capital Stock)</i> ₀₅	0.0182*** (0.0017)
<i>Difference Managerial Education</i> ₀₅	0.0040*** (0.0012)
<i>Difference Manager-Employee Ratio</i> ₀₅	−0.0075 (0.0059)
<i>Difference Unionization Rate</i> ₀₅	0.0065*** (0.0010)
<i>Difference Age</i> ₀₅	0.0005*** (0.0002)
Acquirer-target industry pair FEs	Yes
Acquirer-target state pair FEs	Yes
Observations	10,160,000
Adjusted R ²	0.0001

Notes. This table reports the results from a linear probability model examining the likelihood that a specific firm will acquire a particular target. Coefficient estimates are multiplied by 1,000 to ease interpretation. Each establishment that is eventually acquired is matched to all potential acquiring firms that have management scores. Thus, each observation represents a target establishment-acquirer firm pair. The dependent variable *Became Target* is an indicator variable that is set to one if a firm acquires the establishment. *Difference* indicates that a particular characteristic is calculated by subtracting the establishment-level value of the characteristic from the firm-level value of the same characteristic for each target establishment-acquirer firm pair. Firm-level variables are calculated by averaging each characteristic across all of the firm's establishments. *MP Score* is an establishment's or a firm's management score in 2005. Industry fixed effects (FEs) are based on six-digit NAICS industry codes. Appendix B provides further definitions of the variables. Standard errors in parentheses are heteroscedasticity-robust and clustered by target establishment. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

we estimate the following difference-in-differences OLS regression in changes:

$$\Delta MPS_i = \beta_1 OwnChg_i + \Gamma' \Delta X_i + \Theta' X_{i,2005} + \eta_k + v_s + \varepsilon_i, \quad (2)$$

where ΔMPS_i is the change in the management score of establishment i from 2005 to 2010. We include industry fixed effects (η_k) defined at the six-digit NAICS level and state fixed effects (v_s). Our key variable of interest, $OwnChg_i$ is an indicator variable that is set to one if establishment i is acquired between 2006 and 2010 and zero otherwise. Thus, the coefficient estimate β_1 represents the difference-in-differences estimate comparing the change in the management scores of establishments that are acquired relative to the change in the management scores of establishments that are not acquired over the same time period after accounting for differences between industries and states. We control for a set of establishment-level characteristics X that include changes in the following variables between 2005 and 2010: number of employees, capital stock, average education level across managers, the ratio of the number of managers to nonmanagerial employees, and union membership rate. To account for differences in initial conditions between establishments in 2005, we also control for the 2005 values of each of these variables, as well as the establishment's age and the number of establishments owned by the parent firm.

Because we estimate the model in changes, we difference out any time-invariant differences between establishments, such as differences in industry and geographic location as well as differences in the average level of management practices. However, even though we also control for time-invariant industry and state characteristics with the changes analysis, one could still be concerned that changes in management scores and changes in ownership are correlated with changes in industry and state characteristics, such as changing growth opportunities, product market competition, regulations, and so on. To account for this concern, we also include industry and state fixed effects. Since the panel data set has only two years of observations, and hence one cross section of establishments when the data are in changes, these fixed effects control for any industry and state characteristics that changed between 2005 and 2010. Last, we cluster standard errors by firm to correct for heteroskedasticity and the correlation of standard errors within establishments owned by the same firm (Bertrand et al. 2004).

Table 6 presents results examining the effect of M&A on changes in management practices. Column 1 presents the results from a regression with state fixed effects but no other control variables. The regressions in columns 2–4 sequentially add control variables, with column 2 adding industry fixed effects, column 3

adding establishment-level control variables in changes and establishment age, and column 4 adding the other initial establishment conditions. Across all of these models, the results show that acquired establishments adopt more structured management practices. In terms of economic significance, column 4 shows that the normalized management scores of acquired establishments increase by 0.0254. Given an average increase in management scores of 0.0986 between 2005 and 2010, this additional increase of 0.0254 for acquired establishments is economically significant, representing an increase of 25.8% ($= 0.0254/0.0986$) relative to the average change in management scores. If we compare this 0.0254 increase to the standard deviation of management scores of 0.148 with an average initial value of 0.591, then this increase represents an increase of 17.2% ($= 0.0254/0.148$) of a standard deviation and 4.3% ($= 0.0254/0.591$) relative to the mean.

In column 5, we explore how changes in management scores between 2005 and 2010 are affected by when the establishment is acquired during this time period. Various frictions and adjustment costs can cause management practices to not adjust immediately following an acquisition. For example, prior work suggests that restructuring after a merger can take time, merging workforces can be a lengthy and complex task, and other tasks associated with the merger can draw managements' attention away, resulting in the realization of synergies and changes over a length of time (e.g., Berger et al. 1998, Focarelli and Panetta 2003). A delayed response would also be consistent with Bloom and Van Reenen (2007, p. 1357), who posit that “frictions will slow down the adoption of best management practices. Even if a new management practice were a purely technological innovation, we would expect it to take time to spread.” Therefore, one might predict that acquired establishments will adopt more structured management practices if the acquiring firms have a longer time to make these changes. Thus, establishments acquired closer to 2005 could have larger increases in management scores.

To test this prediction, we decompose the ownership change variable into two indicator variables. *Ownership Change (06–08)* is an indicator variable that is set to one if the establishment is acquired between 2006 and 2008 and zero otherwise. *Ownership Change (09–10)* is an indicator variable that is set to one if the establishment is acquired between 2009 and 2010 and zero otherwise. The results in column 5 are partially consistent with the prediction. Establishments acquired between 2006 and 2008 have an increase in management scores of 0.026, whereas establishments acquired in 2009 or 2010 have an increase of 0.015. However, these values are not statistically different at conventional levels (p -value = 0.17).

Table 6. Changes in Management Practices Following an Acquisition

	Δ MP Score				
	(1)	(2)	(3)	(4)	(5)
<i>Ownership Change</i>	0.0254*** (0.0048)	0.0268*** (0.0045)	0.0251*** (0.0043)	0.0247*** (0.0043)	
Δ Ln(<i>Employment</i>)			0.0072** (0.0028)	0.0097*** (0.0030)	0.0096*** (0.0031)
Δ Ln(<i>Capital Stock</i>)			0.0129*** (0.0036)	0.0124*** (0.0039)	0.0126*** (0.0040)
Δ <i>Managerial Education</i>			0.0553*** (0.0022)	0.0538*** (0.0022)	0.0539*** (0.0022)
Δ <i>Manager-Employee Ratio</i>			−0.0041 (0.0026)	−0.0048** (0.0021)	−0.0048** (0.0021)
Δ <i>Unionization Rate</i>			−0.0010 (0.0041)	−0.0005 (0.0041)	−0.0004 (0.0041)
<i>Age</i> ₀₅			<0.0001 (0.0001)	−0.0002 (0.0001)	−0.0002 (0.0001)
<i>Ln</i> (# <i>Parent Estabs</i>) ₀₅				0.0021** (0.0010)	0.0021** (0.0010)
<i>Ln</i> (<i>Employment</i>) ₀₅				0.0050** (0.0021)	0.0050** (0.0021)
<i>Ln</i> (<i>Capital Stock</i>) ₀₅				−0.0007 (0.0017)	−0.0006 (0.0017)
<i>Managerial Education</i> ₀₅				−0.0024*** (0.0009)	−0.0024*** (0.0009)
<i>Manager-Employee Ratio</i> ₀₅				0.0018 (0.0013)	0.0018 (0.0013)
<i>Unionization Rate</i> ₀₅				0.0009 (0.0007)	0.0009 (0.0007)
<i>Ownership Change</i> (06-08)					0.0259*** (0.0051)
<i>Ownership Change</i> (09-10)					0.0146** (0.0070)
Industry FEs	No	Yes	Yes	Yes	Yes
State FEs	Yes	Yes	Yes	Yes	Yes
Observations	14,000	14,000	14,000	14,000	14,000
Adjusted R^2	0.0061	0.0456	0.1314	0.1330	0.1326

Notes. This table reports the results from OLS regressions examining the effect of mergers and acquisitions on changes in management practices. The dependent variable Δ MP Score is the change in an establishment's management score from 2005 to 2010. *Ownership Change* is an indicator variable that is set to one if an establishment is acquired between 2006 and 2010 and zero otherwise. *Ownership Change* (06-08) is an indicator variable that is set to one if an establishment is acquired between 2006 and 2008 and zero otherwise. *Ownership Change* (09-10) is an indicator variable that is set to one if an establishment is acquired between 2009 and 2010 and zero otherwise. Δ indicates that variables are measured in changes from 2005 to 2010. Industry fixed effects (FEs) are based on six-digit NAICS industry codes. Appendix B provides further definitions of the variables. Standard errors in parentheses are heteroscedasticity-robust and clustered by firm. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

In our analyses, our primary measure of management practices is an establishment's normalized management score aggregated across the three categories of practices—monitoring, production target, and incentive practices. Table 7 examines the effect of M&A on each of these categories of management practices separately. We find significant increases in the management scores in each of the three categories of management practices after an acquisition, and the coefficient

estimates on *Ownership Change* are comparable in magnitude across the models. These results imply that establishments adopt more structured management practices across the board and that one specific category does not drive our findings. In Table A3 in the online appendix, we examine the effect of M&A on each of the 16 management practice questions separately. We find that all 16 management practices become more structured following an acquisition,

Table 7. Changes in Monitoring, Production Target, and Incentive Practices

	Δ MP Monitoring Score	Δ MP Target Score	Δ MP Incentive Score
	(1)	(2)	(3)
<i>Ownership Change</i>	0.0233*** (0.0059)	0.0257*** (0.0048)	0.0269*** (0.0068)
Controls	Yes	Yes	Yes
Industry FEs	Yes	Yes	Yes
State FEs	Yes	Yes	Yes
Observations	14,000	14,000	14,000
Adjusted R^2	0.1257	0.0686	0.0514

Notes. This table reports the results from OLS regressions examining the effect of mergers and acquisitions on changes in management practices. The dependent variables in columns 1–3 are as follows: Δ MP Monitoring Score is the change in an establishment's management score from 2005 to 2010 based on the six questions on monitoring practices; Δ MP Target Score is the change in an establishment's management score from 2005 to 2010 based on the two questions on production target practices; Δ MP Incentive Score is the change in an establishment's management score from 2005 to 2010 based on the eight questions on incentive practices. *Ownership Change* is an indicator variable that is set to one if an establishment is acquired between 2006 and 2010 and zero otherwise. Industry fixed effects (FEs) are based on six-digit NAICS industry codes. Control variables are the same as those in column 4 of Table 6. Appendix B provides further definitions of the variables. Standard errors in parentheses are heteroscedasticity-robust and clustered by firm. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

with 14 of these showing a statistically significant increase in the normalized value.

Taken together, the results presented so far suggest that firms with more structured management practices tend to acquire establishments with relatively less structured management practices. Following the acquisition, the acquirers adopt more structured management practices at the target establishments to realize gains. We test whether this postacquisition adoption of more structured management practices translates into better performance in Section 3.6.

3.3. Convergence in Management Practices Between Acquirers and Targets

Whereas the results in Table 6 show that acquired establishments adopt more structured management practices, we also examine the extent to which the management practices of the target establishment converge to those of the acquiring firm in two ways. First, we examine whether the difference between the acquirer's management score and the target's management score shrinks after the acquisition. Second, we examine whether a target's management practices change in a way that makes them look more like the acquirer's management practices using a measure of distance based on cosine similarity. Table 8 presents the results of these analyses.

For these tests, we create a panel data set in which the unit of observation is a target-acquiring firm pair with observations in 2005 and 2010, and we focus on the subsample of establishments that are eventually acquired. To test the first question, the dependent variable in columns 1 and 2 is the difference between the acquiring firm's average management score and

the acquired establishment's management score in year t (i.e., 2005 or 2010). To test the second question, we set the dependent variable in columns 3 and 4 to the cosine similarity between the acquiring firm's 16 management practices and the acquired establishment's 16 management practices in year t (i.e., 2005 or 2010).¹⁰

The key variable of interest is *Year 2010*, which is an indicator variable that equals one in 2010 and zero otherwise. Because the sample includes only establishments that are acquired between 2006 and 2010, the coefficient estimates on *Year 2010* in columns 1 and 2 represent the difference between an acquirer's management score and the acquired establishment's management score in 2010 relative to this difference in 2005. A negative coefficient estimate would indicate that this difference shrinks, implying that the acquired establishment's management score converges to that of the acquirer. In columns 3 and 4, the coefficient estimate on *Year 2010* represents the distance between an acquirer's management practices and the acquired establishment's management practices in 2010 relative to this distance in 2005. A positive coefficient estimate would indicate that an acquired establishment's management practices become more similar to those of the acquirer after the acquisition. Control variables are the same as those in the gravity model in Table 5 and include differences between each acquiring firm's characteristic and the same characteristic of the acquired establishment. We also include target establishment fixed effects in order to compare differences in management practices from 2005 to 2010 within the same establishment.

Overall, the results show that, in addition to the management practices at acquired establishments

Table 8. Convergence in Management Practices Between Acquirers and Targets

	Difference MP Score		Cosine Similarity MP Score	
	(1)	(2)	(3)	(4)
Year 2010	−0.0226*** (0.0072)	−0.0205*** (0.0070)	0.0498*** (0.0045)	0.0493*** (0.0041)
Difference Ln(Employment)		0.0031 (0.0096)		0.0050 (0.0062)
Difference Ln(Capital Stock)		0.0197 (0.0145)		0.0037 (0.0118)
Difference Managerial Education		0.0653*** (0.0077)		−0.0340*** (0.0045)
Difference Manager-Employee Ratio		−0.0053 (0.0278)		0.0026 (0.0139)
Difference Unionization Rate		−0.0009 (0.0111)		0.0034 (0.0088)
Establishment FEs	Yes	Yes	Yes	Yes
Observations	2,000	2,000	2,000	2,000
Adjusted R ²	0.3687	0.4346	0.1693	0.2259

Notes. This table reports the results from OLS regressions examining whether the difference between an acquiring firm’s management score and the target establishment’s management score converges after the acquisition. The sample is restricted to the subsample of establishments that are eventually acquired by firms with data on management practices. The sample is a panel data set in which the unit of observation is a target-acquiring firm pair with observations in 2005 and 2010. *Difference* indicates that a particular characteristic is calculated by subtracting the target establishment’s value of the characteristic from the acquirer’s value of the same characteristic for each target-acquirer pair. The dependent variable *Difference MP Score* in columns 1 and 2 is the difference between the acquiring firm’s management score and the acquired establishment’s management score in year t (i.e., 2005 or 2010). The dependent variable *Cosine Similarity MP Score* in columns 3 and 4 is the distance between the acquiring firm’s management score and the acquired establishment’s management score in year t (i.e., 2005 or 2010). Endnote 9 defines the calculation of this variable in detail. *Year 2010* is an indicator variable that is set to one in 2010 and zero otherwise. Acquirer-level variables are calculated by averaging each characteristic across all of the firm’s establishments. Industry fixed effects (FEs) are based on six-digit NAICS industry codes. Appendix B provides further definitions of the variables. Standard errors in parentheses are heteroscedasticity-robust and clustered by target establishment. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

becoming more structured on average, the target’s management practices converge to those of the acquirer’s incumbent establishments. Further, the results using the measure of cosine similarity between a target’s and acquirer’s management practices show that the management practices of targets look more similar to those of the acquirer after the acquisition, suggesting that acquirers apply their own management style to the newly acquired establishments.

3.4. Threats to Identification

Our difference-in-differences research design controls for any time-invariant firm- and establishment-level characteristics as well as any time-varying industry and state characteristics, because we include industry and state fixed effects in the two-period change regressions. As a result, the primary identification threat that would not allow us to attribute the changes in the management practices of acquired establishments to the acquisition event is if these changes are a continuation of a pre-existing trend. If acquirers tend to purchase establishments that are

already adopting more structured management practices, then the adoption of more structured practices following an acquisition could simply be a continuation of this trend and not a result of the acquisition.

To investigate this alternative explanation, the ideal test would be to examine whether changes in an establishment’s management score predict the likelihood that the establishment is acquired between 2006 and 2010. However, we do not have data on management practices before 2005. Therefore, we conduct an out-of-sample test. Specifically, we examine whether changes in the management scores of establishments between 2005 and 2010 predict whether the establishments are acquired between 2011 and 2014.

Table 9 presents the results of this analysis. The dependent variable is an indicator variable that is set to one if the establishment is acquired between 2011 and 2014. The variable of interest is $\Delta MP Score_{05-10}$, which is the change in an establishment’s management score from 2005 to 2010. Column 1 controls for establishment employment, productivity, and age as of 2010. Column 2 further includes the same set of

Table 9. Changes in Management Practices and the Likelihood of Becoming a Target

	<i>Became Target₁₁₋₁₄</i>		
	(1)	(2)	(3)
$\Delta MP Score_{05-10}$	0.0093 (0.0200)	0.0119 (0.0216)	−0.0116 (0.0259)
$Ln(Employment)_{10}$	0.0029 (0.0041)	−0.0016 (0.0076)	−0.0128 (0.0104)
TFP_{10}	0.0075 (0.0054)	0.0075 (0.0056)	0.0068 (0.0075)
$MP Score_{05}$		−0.0109 (0.0296)	−0.0370 (0.0360)
Other controls	Yes	Yes	Yes
State FEs	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes
Observations	14,000	14,000	10,000
Adjusted R^2	0.4957	0.4960	0.5535

Notes. This table reports the results from linear probability models examining the effect of changes in an establishment's management score on the likelihood it is acquired. Columns 1 and 2 use the full sample of establishments, whereas column 3 restricts the sample to establishments that have survived through 2014. The dependent variable *Became Target₁₁₋₁₄* is an indicator variable that is set to one if an establishment is acquired between 2011 and 2014. $\Delta MP Score_{05-10}$ is the change in an establishment's management score from 2005 to 2010. Other control variables are the same as those in column 4 of Table 6, except that we replace establishment age in 2005 with establishment age in 2010. Appendix B provides further definitions of the variables. Standard errors in parentheses are heteroscedasticity-robust and clustered by firm. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

control variables from our main model specification in column 4 of Table 6. Whereas we only require that establishments have survived until 2011 to enter the samples in columns 1 and 2, in column 3 we further require the establishments to have survived through 2014 to reduce biases introduced by establishments closing instead of being sold. Overall, the results in all three columns show a statistically insignificant relation between changes in an establishment's management practices and the likelihood that the establishment is subsequently acquired. This result suggests that the changes we observe in an establishment's management practices following an acquisition are likely due to the acquisition event and not just a continuation of a pre-existing trend.

Whereas the prior test attempts to address concerns related to pre-trends, we next conduct a test similar to that in Seru (2014), which exploits announced but withdrawn mergers to generate exogenous variation in the change in management practices of acquired establishments. For this test, we limit the sample to only targeted establishments, creating a treatment group consisting of announced acquisitions that are completed and a control group consisting of announced acquisitions that are subsequently withdrawn. Focusing on only targeted establishments helps alleviate the

concern that the observed changes in management practices following an acquisition are due to selection effects and differences between targeted and non-targeted establishments.

To implement this test, we start with our base sample of 14,000 establishments. We obtain deal announcements from the Securities Data Corporations (SDC) M&A database and fuzzy match announced target names to establishment names in the Census.¹¹ We match roughly 300 establishments to a target name in the SDC database. Of these 300 establishments, the announced merger is completed for about 200 establishments and withdrawn for about 100 establishments (values are rounded per Census disclosure rules). Using this restricted sample, we rerun the regression from column 4 of Table 6 and present the results in Table 10. Due to the restricted sample, the coefficient estimate on *Ownership Change* represents how much management scores change in establishments that are acquired relative to the change in management scores of targeted establishments that are not subsequently acquired. Consistent with our full sample results, the results show that establishments adopt more structured management practices following completed acquisitions. The results also show that monitoring, production target, and incentive scores increase following completed acquisitions, but the increase is statistically significant for only the monitoring and target practices.

3.5. Cross-Sectional Analyses

What characteristics are associated with acquired establishments adopting more structured management practices? If our results are driven by acquirers adopting more structured practices at target establishments to realize synergies, then we should find that acquirers are more likely to make these changes when they have a greater incentive and ability to do so. We test this prediction by exploiting cross-sectional variation in the effect of M&A on changes in an establishment's management practices and present the results of this analysis in Table 11.

First, we examine whether acquirers with more structured management practices adopt more structured practices at target establishments. Acquirers endowed with more structured management practices have the knowledge, skills, and resources that they can use to apply these practices at acquired establishments. To test this prediction, we set the indicator variable *High Acq. MP Score₀₅* to one if the acquirer's management score in 2005 is in the top quartile and zero otherwise. We exclude the main effect of *High Acq. MP Score₀₅*, because this variable can only take a value of one for acquired establishments, and therefore the interaction term absorbs the main effect. Consistent with the prediction, the results

Table 10. Changes in Management Practices: Using Withdrawn M&A Deals as Control Group

	Δ MP Score	Δ MP Monitoring Score	Δ MP Target Score	Δ MP Incentive Score
	(1)	(2)	(3)	(4)
<i>Ownership Change</i>	0.0630** (0.0295)	0.0750** (0.0350)	0.0686** (0.0344)	0.0206 (0.0650)
Controls	Yes	Yes	Yes	Yes
Industry FEs	Yes	Yes	Yes	Yes
State FEs	Yes	Yes	Yes	Yes
Observations	300	300	300	300
Adjusted R^2	0.2643	0.2787	0.1595	0.0756

Notes. This table reports the results from OLS regressions examining the effect of mergers and acquisitions on changes in management practices. The sample includes only establishments that could be matched to an announced acquisition in the SDC database. The dependent variables in columns 1–4 are *MP Score*, *MP Monitoring Score*, *MP Target Score*, and *MP Incentive Score*, which are the normalized management scores based on all of the management practices questions and the questions on monitoring, production target, and incentive practices, respectively. Δ indicates that variables are measured in changes from 2005 to 2010. *Ownership Change* is an indicator variable that is set to one if an establishment was targeted between 2006 and 2010 and the deal was completed and equals zero if an establishment was targeted between 2006 and 2010 but the deal was withdrawn. Control variables are the same as those in column 4 of Table 6. Industry fixed effects (FEs) are based on six-digit NAICS industry codes. Appendix B provides further definitions of the variables. Standard errors in parentheses are heteroscedasticity-robust and clustered by firm. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

in column 1 imply that, relative to acquirers with lower management scores, acquirers with higher management scores adopt even more structured management practices at target establishments.

Second, we test whether an acquired establishment adopts more structured management practices when the acquirer has a greater ability to influence organizational change after it acquires the establishment. The extent to which an organization has centralized or decentralized decision-making authority can have a large effect on organizational change. A centralized organization is one in which decision control is concentrated in higher levels, such as in organizations where establishment managers or other upper-level managers have sole decision-making authority. Whereas there are pros and cons associated with centralized and decentralized structures, a common theme is that centralized structures can be more efficient regarding business decisions in terms of how quick decisions can be made and changes implemented (e.g., Wally and Baum 1994, Baum and Wally 2003). Because there are fewer people involved in the decision-making process in a centralized structure, there are likely fewer conflicts, a smaller need for information sharing and consensus seeking, and less effort spent on time-consuming negotiations and other political behaviors, all of which promote faster decisions and changes.

To test this prediction, we utilize two other questions in the 2010 MOPS. The first question asks, “How many layers of direct reports were there in this establishment from the factory floor to the plant manager?” When there are fewer such layers, managers

have more direct information and a greater ability to directly delegate tasks and promote changes. To indicate instances of a flatter organizational structure, we set the indicator variable *Reporting Layers* to one if the establishment’s response is in the bottom quartile and zero otherwise. The second question asks, “Who prioritized or allocated tasks to production workers?” Responses include (i) only managers, (ii) mostly managers, (iii) managers and production workers jointly, (iv) mostly production workers, (v) only production workers, and (vi) other. To indicate instances when decision-making authority is concentrated among managers, we set the indicator variable *Centralized* to one if the response is only managers and zero otherwise.¹² We measure both of these variables using the establishment’s responses as of 2005. The results in columns 2 and 3 of Table 11 are consistent with our prediction. Acquired establishments adopt even more structured management practices when managers have a greater ability to make decisions and influence organizational change.

Third, we examine whether acquired establishments are more likely to adopt more structured management practices if they face greater product market competition. More intense product market competition serves as a disciplinary device and mitigates managerial slack (e.g., Giroud and Mueller 2010), giving acquirers greater incentives to improve economic efficiency. In addition, if firms are acquiring establishments with less structured management practices, then acquirers could realize larger gains by adopting more structured practices at targets in more

Table 11. Cross-Sectional Variation in Changes in Management Practices

	$\Delta MP Score$				
	(1)	(2)	(3)	(4)	(5)
<i>Ownership Change</i>	0.0210*** (0.0039)	0.0145*** (0.0054)	0.0202*** (0.0046)	0.0171*** (0.0038)	0.0173*** (0.0049)
<i>Ownership Change</i> $\times$ <i>High Acq. MP Score</i> ₀₅	0.0148** (0.0073)				
<i>Ownership Change</i> $\times$ <i>Reporting Layers</i> ₀₅		0.0224*** (0.0074)			
<i>Reporting Layers</i> ₀₅		−0.0171*** (0.0028)			
<i>Ownership Change</i> $\times$ <i>Centralized</i> ₀₅			0.0173** (0.0084)		
<i>Centralized</i> ₀₅			0.0257*** (0.0029)		
<i>Ownership Change</i> $\times$ <i>Low Industry HHI</i>				0.0187** (0.0077)	
<i>Ownership Change</i> $\times$ <i>Same Industry</i> ₀₅					0.0135** (0.0064)
Controls	Yes	Yes	Yes	Yes	Yes
Industry FEs	Yes	Yes	Yes	Yes	Yes
State FEs	Yes	Yes	Yes	Yes	Yes
Observations	14,000	14,000	14,000	14,000	14,000
Adjusted R ²	0.1333	0.1359	0.1415	0.1133	0.1333

Notes. This table reports the results from OLS regressions examining the effect of mergers and acquisitions on changes in management practices. The dependent variable $\Delta MP Score$ is the change in an establishment's management score from 2005 to 2010. *Ownership Change* is an indicator variable that is set to one if an establishment is acquired between 2006 and 2010 and zero otherwise. *High Acq. MP Score*₀₅ is an indicator variable that is set to one if the acquirer's management score is in the top quartile and zero otherwise (it is set to zero for establishments that are not acquired). *Reporting Layers*₀₅ is an indicator variable that is set to one if an establishment's response to how many layers of direct reports there are between the factory floor and the establishment's managers is in the bottom quartile in 2005 and zero otherwise. *Centralized*₀₅ is an indicator variable that is set to one if an establishment's response to who prioritizes and allocates tasks to production workers is only managers in 2005 and zero otherwise. *Low Industry HHI* is an indicator variable that is set to one if the establishment's average industry HHI between 2002 and 2007 is in the bottom quartile and zero otherwise. *Same Industry*₀₅ is an indicator variable that is set to one if the target and the acquirer are in the same three-digit NAICS industry in 2005 and zero otherwise (it is set to zero for establishments that are not acquired). Control variables are the same as those in column 4 of Table 6. Appendix B provides further definitions of the variables. Industry fixed effects (FEs) are based on six-digit NAICS industry codes (columns 1–3 and 5) and three-digit NAICS industry codes (column 4). Standard errors in parentheses are heteroscedasticity-robust and clustered by firm. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

competitive industries. To test this prediction, we measure the degree of product market competition that an establishment faces using the sales-based weighted Herfindahl-Hirschman Index (HHI) of all establishments in its three-digit NAICS industry code that are in the Census Bureau's ASM. In census years (i.e., years ending in "2" and "7"), the ASM surveys all establishments and is also referred to as the Census of Manufacturers in these years. HHI measures based on the entire universe of manufacturing establishments provide a more accurate metric of product market competition and avoid the limitations associated with calculating industry concentration with only public firms (Ali et al. 2009). We measure industry concentration as each industry's average HHI between 2002

and 2007. Lower HHI values indicate more intense competition. Hence, we set the indicator variable *Low Industry HHI* to one if the establishment's industry HHI is in the bottom quartile of the sample and zero otherwise.¹³ We exclude the main effect for *Low Industry HHI*, because it is absorbed by the industry fixed effects. Consistent with greater product market competition incentivizing acquirers to adopt more structured management practices, column 4 shows a larger increase in the management scores of acquired establishments that are in more competitive industries.

Last, acquired establishments are more likely to adopt more structured management practices when targets and acquirers have complementary operations, such as when both operate in the same industry.

When the operations of the target and the acquirer are similar, the acquiring firm presumably already possesses the comparative advantage and expertise needed to operate in the target's industry (e.g., Maksimovic and Phillips 2001, Frésard et al. 2017), making it easier for the acquirer to apply its own management style to the newly acquired establishment. To test this prediction, we set the indicator variable *Same Industry*₀₅ to one if the acquired establishment and at least one of the acquiring firm's establishments are in the same three-digit NAICS industry code in 2005 and zero otherwise. Similar to the regression in column 1, we exclude the main effect for *Same Industry*₀₅, because this variable can only take a value of one for acquired establishments, and therefore the interaction term absorbs the main effect. The results in column 5 are consistent with the prediction that acquired establishments adopt even more structured management practices when an acquirer and a target have complementary operations.

3.6. Changes in Establishment Performance

Is adopting more structured management practices following an acquisition associated with improvements in performance? To test this question, we estimate the following difference-in-differences OLS regression in changes with a continuous treatment effect:

$$\Delta y_i = \beta_1 \text{OwnChg}_i + \beta_2 \text{OwnChg}_i \times \Delta \text{MPS}_i + \beta_3 \Delta \text{MPS}_i + \Gamma' \Delta X_i + \Psi' X_{05} + \eta_k + \nu_s + \varepsilon_i, \quad (3)$$

where Δy_i is the change in one of five measures of an establishment's performance from 2005 to 2011: total factor productivity, the natural logarithm of value added per employee, the natural logarithm of value added per production worker hour, profit margin (value added scaled by total value of shipments), and return on capital (value added scaled by beginning-of-year capital stock). Total factor productivity is the portion of an establishment's output not explained by the traditional sources of inputs used in production, such as capital, labor, and materials (formula defined in Appendix B). Value added is the establishment's value of shipments minus its labor and material costs. Although we know the exact year when an establishment was acquired, we only know whether an establishment's management practices changed at some point between 2006 and 2010 (2005 is the base year). To ensure that we measure performance after the ownership change and the change in management practices, we focus on changes in measures of performance from 2005 to 2011.¹⁴

*OwnChg*_{*i*} is an indicator variable that is set to one if establishment *i* is acquired between 2006 and 2010 and zero otherwise. Like our previous analyses, ΔMPS_i is the

change in establishment *i*'s management score between 2005 and 2010. Our coefficient of interest is β_2 , which represents how much the performance of an establishment changes following an acquisition conditional on the change in its management practices. Thus, positive values of β_2 indicate that the adoption of more structured management practices in acquired establishments is associated with larger improvements in performance. We control for the same set of variables *X* used in our main model specification in column 4 of Table 6. We include industry (η_k) and state (ν_s) fixed effects to control for any industry and state characteristics that changed between 2005 and 2011. We cluster standard errors by firm.

Overall, the results in Table 12 suggest that the adoption of more structured management practices following an acquisition is associated with improvements in performance. Across the five columns, we find statistically significantly larger improvements in four out of five measures of performance in acquired establishments that adopt more structured management practices. For example, the coefficient estimate on *Ownership Change* in column 1 implies that for establishments that do not change their management practices, productivity does not change post acquisition. In contrast, the coefficient estimate on the interaction term implies that for establishments that increase their management scores by one standard deviation, productivity increases by an additional 3.3% following the acquisition ($= 0.2507 \times 0.1305$). To put this magnitude into perspective, prior work finds increases in postmerger productivity between 0.4% and 2.4% (e.g., Maksimovic and Phillips 2001, Schoar 2002, Li 2013), suggesting that the improvement in productivity attributable to adopting more structured management practices is economically significant. Joint significance tests also indicate that productivity increases after an acquisition and after establishments adopt more structured management practices.

The results in columns 2–4 show a similar pattern. Following acquisitions in which establishments increase their management scores by one standard deviation, value added per employee, value added per worker hour, and profit margins increase by an additional 3.13, 4.19, and 1.16 percentage points, respectively. Whereas joint significance tests for $\Delta \text{MP Score}_{05-10} + \text{Ownership Change} \times \Delta \text{MP Score}_{05-10} = 0$ indicate that adopting more structured management practices is associated with improvement in these other performance metrics (except return on capital), the joint tests for $\text{Ownership Change} + \text{Ownership Change} \times \Delta \text{MP Score}_{05-10} = 0$ indicate statistically insignificant improvements in performance following an acquisition.¹⁵

In Table 13, we focus on total factor productivity as our measure of performance and examine how productivity

Table 12. Changes in Performance Following an Acquisition

	ΔTFP_{05-11}	$\Delta \ln(VA/Emp)_{05-11}$	$\Delta \ln(VA/Emp\ Hr)_{05-11}$	ΔPM_{05-11}	ΔROC_{05-11}
	(1)	(2)	(3)	(4)	(5)
<i>Ownership Change</i>	−0.0429 (0.0301)	−0.1298*** (0.0263)	−0.1092*** (0.0277)	−0.0153 (0.0100)	−0.4123** (0.1735)
<i>Ownership Change</i> × $\Delta MP\ Score_{05-10}$	0.2507** (0.1171)	0.2396* (0.1398)	0.3211** (0.1476)	0.0886* (0.0535)	0.4607 (0.9236)
$\Delta MP\ Score_{05-10}$	−0.0311 (0.0480)	0.1763*** (0.0622)	0.1242* (0.0657)	0.0025 (0.0238)	−0.1492 (0.4110)
Controls	Yes	Yes	Yes	Yes	Yes
Industry FEs	Yes	Yes	Yes	Yes	Yes
State FEs	Yes	Yes	Yes	Yes	Yes
Observations	14,000	14,000	14,000	14,000	14,000
Adjusted R^2	0.1187	0.0608	0.0446	0.0211	0.0828

Notes. This table reports the results from OLS regressions examining the effect of mergers and acquisitions on changes in establishment performance over the years 2005 to 2011. The dependent variables measured as the change in their values from 2005 to 2011 are as follows: *TFP* is establishment total factor productivity estimated following Foster et al. (2016); $\ln(VA/Emp)$ is the natural logarithm of establishment value added scaled by the establishment's number of employees; $\ln(VA/Emp\ Hr)$ is the natural logarithm of establishment value added scaled by the establishment's total number of production worker hours; *PM* is establishment value added (value of shipments minus labor and material costs) scaled by total value of shipments; *ROC* is establishment value added scaled by beginning of year capital stock. $\Delta MP\ Score_{05-10}$ is the change in an establishment's management score from 2005 to 2010. *Ownership Change* is an indicator variable that is set to one if an establishment is acquired between 2006 and 2010 and zero otherwise. Control variables are the same as those in column 4 of Table 6. Appendix B provides further definitions of the variables. Dependent variables are winsorized at their 1st and 99th percentiles. Standard errors in parentheses are heteroscedasticity-robust and clustered by firm. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

changes following an acquisition conditional on changes in monitoring, production target, and incentive practices separately.¹⁶ Overall, whereas the results are weaker, we find larger improvements in productivity after an acquisition when firms adopt more structured monitoring, production target, and incentive practices. However, the improvements in productivity are statistically significant only at the 10% level for the monitoring and incentive practices.

In Table A4 in the online appendix, we further examine how productivity changes following an acquisition conditional on changes in each of the 16 management practice questions separately. We find larger improvements in productivity after an acquisition when firms adopt more structured monitoring practices for five out of six questions, with four of these being statistically significant. More structured target practices are associated with larger postacquisition improvements in productivity for one out of two questions, but it is not statistically significant (p -value = 0.11). Last, we find larger improvements in productivity after an acquisition when firms adopt more structured incentive practices for six out of eight questions, but only three of these are statistically significant. Overall, whereas these results are generally consistent with those using aggregated management scores, breaking out the changes in management practices by individual question appears to produce noisy estimates.

In Table A5 in the online appendix, we conduct a similar analysis as that in column 5 of Table 6, in which we examine how postacquisition changes in

establishment performance are affected by when the establishment is acquired. If changes in performance are due to postmerger changes in management practices, then acquisitions made earlier in the sample period would likely have larger effects. To conduct this analysis, we rerun the same regressions as in Table 12 but replace the indicator variable for whether an establishment was acquired at any point between 2006 and 2010 with two separate variables indicating whether the establishment was acquired between 2006 and 2008 (*Ownership Change (06-08)*) or acquired between 2009 and 2010 (*Ownership Change (09-10)*). Similar to the results in Table 6, we find that adopting more structured management practices after an acquisition is associated with statistically significant larger improvements in four out five of the performance measures if the acquisition occurred between 2006 and 2008, but there are no statistically significant changes in performance if the acquisition occurred between 2009 and 2010.

3.7. Concerns About Recall Bias, Sampling Noise, and Response Manipulation

To create a longitudinal data set, each respondent was asked about her or his establishment's management practices in 2005. As a result, the longitudinal component of the 2010 MOPS relies on the recall ability of the respondents, and there could be concerns about recall bias and therefore about the quality of the responses for 2005. There is also a potential concern about sampling noise if, for example, larger establishments have more senior managers filling out the

Table 13. Changes in Productivity Following an Acquisition by Category of Management Practices

	ΔTFP_{05-11}		
	(1)	(2)	(3)
<i>Ownership Change</i>	−0.0431 (0.0345)	−0.0182 (0.0252)	−0.0270 (0.0241)
<i>Ownership Change</i> × ΔMP Monitoring Score ₀₅₋₁₀	0.1644* (0.0888)		
ΔMP Monitoring Score ₀₅₋₁₀	−0.0511 (0.0327)		
<i>Ownership Change</i> × ΔMP Target Score ₀₅₋₁₀		0.0368 (0.0588)	
ΔMP Target Score ₀₅₋₁₀		0.0210 (0.0224)	
<i>Ownership Change</i> × ΔMP Incentive Score ₀₅₋₁₀			0.2072* (0.1069)
ΔMP Incentive Score ₀₅₋₁₀			−0.0126 (0.0465)
Controls	Yes	Yes	Yes
Industry FEs	Yes	Yes	Yes
State FEs	Yes	Yes	Yes
Observations	14,000	14,000	14,000
Adjusted R ²	0.1187	0.1184	0.1187

Notes. This table reports the results from OLS regressions examining the effect of mergers and acquisitions on changes in establishment total factor productivity over the years 2005 to 2011. The dependent variable ΔTFP_{05-11} in columns 1–3 is the change in establishment total factor productivity estimated following Foster et al. (2016) from 2005 to 2011. ΔMP Monitoring Score₀₅₋₁₀, ΔMP Target Score₀₅₋₁₀, and ΔMP Incentive Score₀₅₋₁₀ are the normalized management scores based on the questions on monitoring, production target, and incentive practices from 2005 to 2010, respectively. *Ownership Change* is an indicator variable that is set to one if an establishment is acquired between 2006 and 2010 and zero otherwise. Control variables are the same as those in column 4 of Table 6. Appendix B provides further definitions of the variables. Dependent variables are winsorized at their 1st and 99th percentiles. Standard errors in parentheses are heteroscedasticity-robust and clustered by firm. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

survey and these managers tend to be more optimistic or pessimistic in evaluating their management practices.

To examine the extent to which these concerns affect our results, we follow Bloom et al. (2013b, 2019) and show in Table A6 in the online appendix that our main result is robust to accounting for a full range of bias and noise controls. In particular, our results are robust to (i) controlling for the respondent's tenure to account for the concern that the respondent filling out the survey was not working at the establishment in 2005 (average tenure is 11.7 years), (ii) controlling for the difference between a respondent's recalled number of employees at her or his establishment in 2005 and 2010 and the actual number of employees reported in the ASM in these years (recall bias is likely greater for respondents that could not recall the correct number of employees), (iii) controlling for whether the respondent completed the survey online or by mail, and (iv) including fixed effects for the respondent's position at the establishment.

A potential concern associated with reporting management practices in this setting is that respondents may try to manipulate their responses by claiming that their

establishment has adopted more structured management practices when an acquisition turns out well but claiming that their establishment has adopted less structured practices when an acquisition turns out poorly. Respondents may have incentives to game their responses in this way if their responses are reviewed by their superiors and are somehow tied to their compensation or career prospects. This effect could explain our finding of larger postacquisition improvements in performance when establishments adopt more structured management practices. Overall, we do not believe that this alternative explanation is likely. The MOPS is distributed as part of the ASM to a predetermined contact person at an establishment. This contact person simply fills out the survey and returns it to the Census Bureau. The responses are generally not distributed to others at the firm. Thus, there does not appear to be any incentive for respondents to try to game their responses.

Nevertheless, if respondents are manipulating their responses, then we might expect differences across types of respondents, because they would face different incentives. For example, an establishment manager

might have more incentives than a CEO to show that management practices changed after an acquisition, because they are accountable to more senior-level managers. In column 5 of Table OA.6, we test this prediction by examining whether the change in management scores following an acquisition is different depending on who responds to the survey (i.e., CEO, district manager, establishment manager, or a non-manager). We do not find much variation in the changes in management scores across different respondents, with the exception that establishment managers tend to actually report smaller postacquisition changes in management practices.

4. Conclusion

Using a novel establishment-level survey data set of management practices, we provide new evidence that the adoption of more structured management practices constitutes a source of value creation in corporate M&A. We find that firms with more structured management practices tend to acquire establishments with less structured management practices, and, following the acquisition, adopt more structured practices at the target establishments. Further, acquired establishments adopt even more structured management practices when acquirers are endowed with more structured management practices, managers have greater decision control, establishments are in more competitive industries, and acquirers and targets have complementary operations. We also show that the adoption of more structured management practices is associated with improvements in performance, as measured by productivity, value added per worker, and profit margins.

Our paper is the first large-scale study to document that the adoption of more structured management practices is a source of value creation in M&A. Our results suggest that some establishments underinvest in management practices related to monitoring, production targets, and incentives and that these establishments could improve their performance by adopting more structured management practices. Thus, a remaining question is if more structured management practices could improve performance, then why is it that all firms do not immediately adopt these practices? Several reasons exist (e.g., Bloom and Van Reenen 2007). First, changing management practices can be costly in terms of time and expenditures, and some firms may view these

costs as exceeding any benefits. Second, implementing more structured management practices can increase the amount of effort required of managers, and entrenched managers may prefer the status quo. Third, various frictions, such as learning effects and adjustment costs, could slow down changes in management practices. One possible reason that we observe changes in management practices following an acquisition is that the acquisition event changes the cost-benefit analysis and could therefore be an ideal time to adopt more structured management practices.

We believe that with the availability of this high-quality data on management practices, understanding how management practices interact with a firm's financial condition and various other decisions could be a fruitful area for future research.

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Appendix A. MOPS Questions

Here, we reproduce the 16 questions used to construct each establishment's management score. We tabulate each question and the management practice type associated with the question, the possible responses to each question, and the normalized value of each response to a score between zero and one. The scoring method in the column labeled *Score* follows Bloom et al. (2013b) and Bloom et al. (2019). For additional question details, the complete 2010 Management and Organizational Practice Survey questionnaire is available at <https://www.census.gov/programs-surveys/mops/technical-documentation/questionnaires.html>.

Q1 Monitoring: What best describes what happened at this establishment when a problem in the production process arose?

Response	Score
No action was taken	0.000
We fixed it but did not take further action	0.333

Appendix A. (Continued)

Q1 Monitoring: What best describes what happened at this establishment when a problem in the production process arose?

Response	Score
We fixed it and took action to make sure that it did not happen again	0.667
We fixed it and took action to make sure that it did not happen again, and had a continuous improvement process to anticipate problems like these in advance	1.000

Q2 Monitoring: How many key performance indicators were monitored at this establishment?

Response	Score
No key performance indicators	0.000
1–2 key performance indicators	0.333
3–9 key performance indicators	0.667
10 or more key performance indicators	1.000

Q3 Monitoring: How frequently were the key performance indicators reviewed by managers at this establishment?

Response	Score
Never	0.000
Yearly	0.167
Quarterly	0.333
Monthly	0.500
Weakly	0.667
Daily	0.833
Hourly or more frequently	1.000

Q4 Monitoring: How frequently were the key performance indicators reviewed by nonmanagers at this establishment?

Response	Score
Never	0.000
Yearly	0.167
Quarterly	0.333
Monthly	0.500
Weakly	0.667
Daily	0.833
Hourly or more frequently	1.000

Q5 Monitoring: Where were the production display boards showing output and other key performance indicators located at this establishment?

Response	Score
We did not have any display boards	0.000
All display boards were located in one place (e.g., at the end of the production line)	0.500
Display boards were located in multiple places (e.g., at multiple stages of the production line)	1.000

Q6 Targets: What best describes the time frame of production targets at this establishment?

Response	Score
No production targets	0.000
Main focus was on short-term (less than one year) production targets	0.333
Main focus was on long-term (more than one year) production targets	0.667
Combination of short-term and long-term production targets	1.000

Q7 Targets: How easy or difficult was it for this establishment to achieve its production targets?

Response	Score
Possible to achieve without much effort	0.000
Only possible to achieve with extraordinary effort	0.250
Possible to achieve with some effort	0.500
Possible to achieve with normal amount of effort	0.750
Possible to achieve with more than normal effort	1.000

Appendix A. (Continued)

Q8 Monitoring: Who was aware of the production targets at this establishment?

Response	Score
Only senior managers	0.000
Most managers and some production workers	0.333
Most managers and most production workers	0.667
All managers and most production workers	1.000

Q9 Incentives: What were nonmanagers' performance bonuses usually based on?

Response	Score
No performance bonuses	0.000
Their company's performance as measured by production targets	0.250
Their establishment's performance as measured by production targets	0.500
Their team or shift performance as measured by production targets	0.750
Their own performance as measured by production targets	1.000

Q10 Incentives: When production targets were met, what percent of nonmanagers at this establishment received performance bonuses?

Response	Score
Production targets not met	0.000
0%	0.200
1%–33%	0.400
34%–66%	0.600
67%–99%	0.800
100%	1.000

Q11 Incentives: What were managers' performance bonuses usually based on?

Response	Score
No performance bonuses	0.000
Their company's performance as measured by production targets	0.250
Their establishment's performance as measured by production targets	0.500
Their team or shift performance as measured by production targets	0.750
Their own performance as measured by production targets	1.000

Q12 Incentives: When production targets were met, what percentage of managers at this establishment received performance bonuses?

Response	Score
Production targets not met	0.000
0%	0.200
1%–33%	0.400
34%–66%	0.600
67%–99%	0.800
100%	1.000

Q13 Incentives: What was the primary way nonmanagers were promoted at this establishment?

Response	Score
Nonmanagers are normally not promoted	0.000
Promotions were based mainly on factors other than performance and ability (for example, tenure or family connections)	0.333
Promotions were based partly on performance and ability, and partly on other factors (for example, tenure or family connections)	0.667
Promotions were based solely on performance and ability	1.000

Q14 Incentives: What was the primary way managers were promoted at this establishment?

Response	Score
Nonmanagers are normally not promoted	0.000
Promotions were based mainly on factors other than performance and ability (for example, tenure or family connections)	0.333
Promotions were based partly on performance and ability, and partly on other factors (for example, tenure or family connections)	0.667
Promotions were based solely on performance and ability	1.000

Appendix A. (Continued)

Q15 Incentives: When was an under-performing nonmanager reassigned or dismissed?	
Response	Score
Rarely or never	0.000
After six months of identifying nonmanager under-performance	0.500
Within six months of identifying nonmanager under-performance	1.000
Q16 Incentives: When was an under-performing manager reassigned or dismissed?	
Response	Score
Rarely or never	0.000
After six months of identifying nonmanager under-performance	0.500
Within six months of identifying nonmanager under-performance	1.000

Appendix B. Variable Definitions

# Parent Estabs	The number of establishments owned by the parent firm.
Age	The number of years since the establishment's initial appearance in the LBD. Firm-level <i>Age</i> is the number of years that the firm's oldest establishment has been in existence in the LBD, which follows Foster et al. (2016).
Capital Stock	The capitalized value of property, plant, and equipment of the establishment calculated using the perpetual inventory method following Foster et al. (2016).
Employment	The number of total employees at the establishment. Firm-level <i>Employment</i> is the sum of the total number of employees at all of the establishments owned by the firm.
Manager-Employee Ratio	The number of managers at the establishment divided by the number of nonmanager employees.
Managerial Education	The normalized responses to the question of what percent of the establishment's managers have a bachelor's degree, with responses ranging from (i) 20% or less to (v) more than 80%.
MP Incentive Score	An establishment's normalized management score based on the questions on incentive practices.
MP Monitoring Score	An establishment's normalized management score based on the questions on monitoring practices.
MP Target Score	An establishment's normalized management score based on the questions on production target practices.
MP Score	An establishment's normalized management score based on all 16 management practice questions. Firm-level <i>MP Score</i> is the average normalized management score across a firm's establishments.
Ownership Change	An indicator variable that is set to one if an establishment is acquired between 2006 and 2010 and zero otherwise.
TFP	An establishment's total factor productivity estimated using the Cobb-Douglas specification following Foster et al. (2016): $TFP_{i,t} = \ln Q_{i,t} - \alpha_K \ln K_{i,t} - \alpha_L \ln L_{i,t} - \alpha_M \ln M_{i,t}$, where Q is real output (value of shipments adjusted by industry deflators), K is capital stock, L is labor input, M is materials, α denotes factor elasticities, the subscript i denotes individual establishments, and the subscript t denotes time. Foster et al. (2016) standardize the computation of capital stock using establishment data with the perpetual inventory method (for further details, see appendix B in Foster et al. (2016)). Firm-level <i>TFP</i> is the average of <i>TFP</i> across all of the firm's establishments.
Unionization Rate	The normalized responses to the question of what percent of the establishment's employees are members of a labor union, with responses ranging from (i) 0% to (vi) more than 80%.

Endnotes

¹ An establishment is a single physical location where business is conducted or where services or industrial operations are performed.

² In giving advice on how to successfully integrate an acquired business, professional groups often discuss the importance of setting clear performance goals and monitoring these goals to ensure that the integration process is on track, such as by developing an M&A scorecard (e.g., Deloitte 2010, Protiviti 2016, PwC 2017, Meyer 2020). These groups also highlight the importance of identifying key employees and properly incentivizing workers in order to successfully integrate a new business.

³ Whereas we conduct the majority of our analyses on establishment-level data, we conduct our tests examining whether there is a relation between management practices and the likelihood that a firm becomes an acquirer on firm-level data. In these firm-level analyses, we calculate the employment-based weighted-average management score across all of a firm's establishments.

⁴ We note that these analyses do not identify exogenous variation in the likelihood that an establishment becomes a target or exogenous changes in the establishment's management practices after the merger.

Thus, whereas our results show that adopting more structured management practices following a merger are associated with larger improvements in the targeted establishment's performance, we cannot conclude that these associations are causal.

⁵ Another related paper is that by Bernstein and Sheen (2016), who show that private equity buyouts of restaurant chains are associated with improvements in store-level operational practices, such as cleanliness, safety, and maintenance.

⁶ A potential concern called *recall bias* arises in surveys when respondents are asked to recall information from an earlier time period. Recall bias in our setting is a concern, because it could affect the quality of the responses for 2005. We address this concern in detail in Section 3.7. Overall, accounting for potential recall bias has very little effect on our results.

⁷ For some questions, respondents can select more than one answer per year. For these questions, we use the average of the normalized answers as the score for the particular question.

⁸ According to the Census Bureau's disclosure policy, observations exceeding 1,000 are rounded to the nearest thousands, and observations between 100 and 1,000 are rounded to the nearest hundreds.

⁹ The ASM panel is a rotating panel of manufacturing establishments, in which establishments with over 250 employees are surveyed with a probability of one and smaller establishments are surveyed with a preassigned probability. A new rotating panel is introduced two years after each U.S. Census (i.e., years ending in “2” and “7”).

¹⁰ Our cosine similarity measure of the distance between the management practices of acquirers and targets equals the dot product of the normalized vectors (by vector length) for the management practices of acquirers and targets ($\frac{T \cdot A}{TA}$). T and A are vectors containing 16 elements that correspond to the 16 management practices questions for the target and acquirer, respectively. Each element contains a value between 0 and 1 that corresponds to the response to the particular survey question.

¹¹ We match the name of the target in the SDC database to the name of the establishment/firm in LBD. We perform a fuzzy name match. We keep exact name matches. For close name matches, we manually determine whether the match is correct and keep those that match. We note that several of the announced SDC deals are partial acquisitions. It is possible that some of the establishments that we match to the SDC database are not the ones that would have been taken over. This measurement error is inherent in this test. One solution is to focus only on deals in which the bidder is going to acquire 100% of the target, but the sample shrinks too much to conduct any meaningful analyses. Overall, this measurement error is unavoidable, and the results should be interpreted accordingly.

¹² We choose how to split our sample in these cross-sectional tests based on the distribution of the variables. We choose a method that ensures that each cell of the interaction is well populated in order to comply with the Census Bureau’s disclosure policy.

¹³ Unlike the other regressions in Table 11, column 4 includes three-digit NAICS industry fixed effects instead of six-digit NAICS industry fixed effects. We use three-digit NAICS fixed effects to be consistent with how we define industry HHI. However, the results are robust to including six-digit NAICS fixed effects.

¹⁴ Whereas this approach guarantees that we capture changes in performance after the acquisition and the change in management practices, we cannot guarantee that a target’s management practices changed after the acquisition with our data. However, our tests showing that changes in management practices do not predict the likelihood that an establishment subsequently becomes a target (Table 9) and that there are larger changes in management practices after completed acquisitions than announced acquisitions that are withdrawn (Table 10) suggest that the acquisition event leads to the changes in management practices. Nevertheless, we cannot rule out this possibility. We have also rerun these changes in performance analyses using longer horizons (i.e., changes from 2005 to 2012, 2005 to 2013, and 2005 to 2014). Whereas most of the coefficient estimates remain positive, the estimates become less precise.

¹⁵ Across the measures of performance in Table 12, the results show that establishment performance tends to worsen, on average, following a merger, which is inconsistent with the findings in prior work (e.g., Li 2011). We conduct additional analyses using the full ASM sample and alternative sample periods to investigate the source of this discrepancy. Overall, the results suggest that it is not something unique about the MOPS sample that drives this negative effect. Rather, it appears that our findings are due to something unique about the period over which we measure performance (i.e., changes in performance from 2005 to 2011). If we conduct the same analysis for each year ending between 2003 and 2014, then the results show a general pattern that, as long as the window over which we measure performance does not overlap with a recession, ownership changes are associated with improvements in performance. When the performance window overlaps with a recession, the results tend to show that ownership changes are associated with worsening performance.

Nevertheless, even if the average effect of an acquisition on an establishment’s performance is negative during our sample period, our results suggest that adopting more structured management practices is associated with larger relative improvements in performance. Thus, our conclusions remain the same (Li 2013).

¹⁶ Whereas we find that the other four measures of performance from columns 2–5 of Table 12 also improve more following an acquisition for establishments that adopted more structured monitoring, production target, and incentive practices, these improvements are not statistically significant at conventional levels.

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